ODE Definitions and Terminology

- An **IVP** is an **ODE** relating $\boldsymbol{y}(t)$ to its derivatives, with an **IC** $\boldsymbol{y}(t_0) = y_0$
  - Highest derivative is **Order**. A **Solution** satisfies ODE and IC on $a < t_0 < b$
  - **Autonomous** no explicit t. **Equilibrium** a constant solution $\boldsymbol{f}(\boldsymbol{y}_*) = \boldsymbol{0}$
  - An ODE **General Solution** contains all solutions with constants determined by ICs.
- **Separable** form $y' = g(t)h(y)$; linear **Standard Form** is $y' + p(t)y = q(t)$
- $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{b}(t)$ is **Linear**.
  $\boldsymbol{A}(t)$ constant is **Constant Coefficient (CC)**. $b = 0$ is **Homogeneous**.
- The general solution of CC homogeneous ODE $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$ is

$$\boldsymbol{y}(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t} \boldsymbol{v}_i$$

  provided $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots \boldsymbol{v}_n$ is a complete LI set of eigenvectors for $\boldsymbol{A}$.
  The constants $\boldsymbol{c}$ satisfy $\boldsymbol{V}\boldsymbol{c} = \boldsymbol{y}(0)$.

Complementary and Particular Solutions for $y' = A(t)y + b(t)$

For a **first-order linear system** the **general solution (GS)** has the form

$$\boxed{\boldsymbol{y}(t) = \boldsymbol{y}_c(t) + \boldsymbol{y}_p(t).}$$

- The **complementary solution** $\boldsymbol{y}_c$ satisfies $\boldsymbol{y}_c' = \boldsymbol{A}(t)\boldsymbol{y}_c$.
  - Contains the arbitrary constants of integration $\boldsymbol{c}$.
  - We know how to compute it for constant coefficients.
- A **particular solution** $\boldsymbol{y}_p$ satisfies $\boldsymbol{y}_p' = \boldsymbol{A}(t)\boldsymbol{y}_p + \boldsymbol{b}(t)$.
  - Any method that finds one such solution is fine.
- $\boldsymbol{y}(0) = \boldsymbol{y}_c(0) + \boldsymbol{y}_p(0) = \boldsymbol{y}_0$. Initial conditions determine the constants $\boldsymbol{c}$.

Constant Forcing

$$y' = \overbrace{\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}}^{A} y + \overbrace{\begin{pmatrix} 3 \\ 0 \end{pmatrix}}^{f}$$

- Try $\boldsymbol{y}_p(t) = \boldsymbol{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ which gives $\boldsymbol{y}_p' = \boldsymbol{0}$ in the ODE

$$\boldsymbol{0} = \boldsymbol{A}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \boldsymbol{f} \implies \boldsymbol{A}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = -\boldsymbol{f} \implies \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}.$$

$$\left[\begin{array}{cc|c} -2 & 1 & -3 \\ 1 & -2 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & -3 & -3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right] \implies \boldsymbol{y}_p(t) = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

- $\boldsymbol{y}(t) = \boldsymbol{y}_p(t) + \boldsymbol{y}_c(t) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_1 e^{-t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \sim \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ as $t \to \infty$
- An IC $\boldsymbol{y}(0) = \boldsymbol{y}_0$ determine the constants $\boldsymbol{c}$ from the linear system

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \boldsymbol{y}_0 - \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Linear Forcing

$$y' = \overbrace{\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}}^{A} y + \overbrace{\begin{pmatrix} 6 \\ -3 \end{pmatrix} t + \begin{pmatrix} 0 \\ 3 \end{pmatrix}}^{f(t)}$$

- Try $\boldsymbol{y}_p(t) = \boldsymbol{a}t + \boldsymbol{b}$ which gives $\boldsymbol{y}_p' = \boldsymbol{a}$ in the ODE and match powers of t

$$\boldsymbol{a} = \boldsymbol{A}\boldsymbol{a}t + \boldsymbol{A}\boldsymbol{b} + \begin{pmatrix} 6 \\ -3 \end{pmatrix} t + \begin{pmatrix} 0 \\ 3 \end{pmatrix} \implies \boldsymbol{A}\boldsymbol{a} = -\begin{pmatrix} 6 \\ -3 \end{pmatrix} \qquad \boldsymbol{A}\boldsymbol{b} = \boldsymbol{a} - \begin{pmatrix} 0 \\ 3 \end{pmatrix}$$

$$\boldsymbol{A}\boldsymbol{a} = -\begin{pmatrix} 6 \\ -3 \end{pmatrix} = \begin{pmatrix} -6 \\ 3 \end{pmatrix} \implies \left[\begin{array}{cc|c} -2 & 1 & -6 \\ 1 & -2 & 3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 0 \end{array}\right] \implies \boldsymbol{a} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

$$\boldsymbol{A}\boldsymbol{b} = \boldsymbol{a} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix} \implies \left[\begin{array}{cc|c} -2 & 1 & 3 \\ 1 & -2 & -3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 1 \end{array}\right] \implies \boldsymbol{b} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

- $\boldsymbol{y}(t) = \boldsymbol{y}_p(t) + \boldsymbol{y}_c(t) = \begin{pmatrix} 3 \\ 0 \end{pmatrix} t + \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_1 e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \sim \begin{pmatrix} 3t \\ 1 \end{pmatrix}$ as $t \to \infty$
- As before initial conditions $\boldsymbol{y}(0) = \boldsymbol{y}_0$ determine the constants.

Quadratic Forcing

$$\boldsymbol{y}' = \overbrace{\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}}^{\boldsymbol{A}} \boldsymbol{y} + \overbrace{\begin{pmatrix} 6 \\ -3 \end{pmatrix} t^2 + \begin{pmatrix} 0 \\ 3 \end{pmatrix} t + \begin{pmatrix} 0 \\ 3 \end{pmatrix}}^{\boldsymbol{f}(t)}$$

- Try $\boldsymbol{y}_p(t) = \boldsymbol{a}t^2 + \boldsymbol{b}t + \boldsymbol{d}$ which gives $\boldsymbol{y}_p' = 2\boldsymbol{a}t + \boldsymbol{b}$ in the ODE and match powers of t

$$2\boldsymbol{a}t + \boldsymbol{b} = \boldsymbol{A}\boldsymbol{a}t^2 + \boldsymbol{A}\boldsymbol{b}t + \boldsymbol{A}\boldsymbol{d} + \begin{pmatrix} 6 \\ -3 \end{pmatrix} t^2 + \begin{pmatrix} 0 \\ 3 \end{pmatrix} t + \begin{pmatrix} 0 \\ 3 \end{pmatrix}$$

$$\boldsymbol{A}\boldsymbol{a} = -\begin{pmatrix} 6 \\ -3 \end{pmatrix} \implies \left[\begin{array}{cc|c} -2 & 1 & -6 \\ 1 & -2 & 3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 0 \end{array}\right] \implies \boldsymbol{a} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

$$\boldsymbol{A}\boldsymbol{b} = 2\boldsymbol{a} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix} \implies \left[\begin{array}{cc|c} -2 & 1 & 6 \\ 1 & -2 & -3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 1 & 0 \end{array}\right] \implies \boldsymbol{b} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}$$

$$\boldsymbol{A}\boldsymbol{d} = \boldsymbol{b} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \end{pmatrix} \implies \left[\begin{array}{cc|c} -2 & 1 & -3 \\ 1 & -2 & -3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 3 \end{array}\right] \implies \boldsymbol{d} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

- $\boldsymbol{y}(t) = \begin{pmatrix} 3t^2 - 3t + 3 \\ 3 \end{pmatrix} + c_1 e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \sim \begin{pmatrix} 3t^2 \\ 3 \end{pmatrix}$ as $t \to \infty$